NAME: _____

DATE: _____

SYSTEM: Generic Admin - Emergency Plan

TASK: Classify an Event and Determine PARs

TASK NUMBER: 1240020502

INITIAL CONDITIONS:

- Salem Units 1 and 2 are at 100% power.
- 23 Station Power Transformer (SPT) is tagged and is 12 hours into a 24 hour scheduled maintenance window.
- An electrical grid disturbance on the 500 kV lines has caused the Unit 2 Main Generator protection circuit to automatically trip the Unit 2 reactor.
- The crew is transitioning from EOP-TRIP-1 to EOP-TRIP-2.
- Following transition to TRIP-2, 24 SPT de-energizes due to actuation of electrical protection relays.
- The 2A Emergency Diesel Generator (EDG) started then tripped on overspeed and could not be reset; and the 2B 4 KV Vital Bus de-energized following a Bus Differential protection trip.
- Minutes later the Primary Operator reports a loud sound from the EDG rooms and 2C EDG trips.

INITIATING CUE:

- Maintenance provides you the following updates:
  - 2A EDG will take 5 hours to repair the bent linkage to the fuel racks.
  - 2B 4 KV Vital Bus has sustained significant damage to the bus bars due to a ground fault.
  - 2C EDG has sustained significant engine damage due to an engine piston failure.
  - 23 SPT needs 8 hours to restore and release tags.
  - 24 SPT sustained an internal fault and will need outside vendor support. Vendor is expected to be on-site in 3.5 hours.
- You are the Emergency Coordinator:
  1. **Classify the emergency event, and**
  2. **Complete the Initial Contact Message Form (ICMF)** .
- MET computer 33 FT. level wind direction is steady from 90 degrees at 2 mph.
- 2R41D, Plant Vent Radiation monitor is reading normal.

*****THIS IS A TIME CRITICALL JPM*****

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made.

Task Standard for Successful Completion:

1. **Classifies the event as <u>GENERAL EMERGENCY</u> (GE) based on <u>SG1.1</u> ≤ 15 mins.**
2. **Completes Attachment 4 Sections I thru V of the ICMF <u>and</u> selects Default PAR (No RPSA) ≤ 15 minutes from event declaration (see ICMF Answer Key).**

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System: Generic Admin - Emergency Plan

Task: Classify an Event and Determine PARs

* #	STEP NO.	STEP (Shaded area denotes Critical Step) (* Critical Step) (# Sequential Critical Step)	STANDARD (Bolded area identifies Task Standard)	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
		Provide the following materials: ▪ Salem Emergency Action Levels (EALs) Flowcharts & Attachments book ▪ Salem Emergency Action Levels (EALs) Technical Basis book ▪ Attachments 1 thru 4 (EP-SA-325-F1/F2/F3/F4)			
	CUE:	**State to the Operator:** **THIS IS A TIME CRITICAL JPM.** **You will be giving time to REVIEW the Initial Conditions. The JPM will START once you have completed your review and acknowledge the Initiating Cue. The time will stop when you have classified the event and submitted the ICMF to the Lead Evaluator.** Fill in the JPM Start Time when the student acknowledges the Initiating Cue. **START TIME**:_____	Operator reviews initial conditions.		

System: Generic Admin - Emergency Plan

Task: Classify an Event and Determine PARs

* #	STEP NO.	STEP (Shaded area denotes Critical Step) (* Critical Step) (# Sequential Critical Step)	STANDARD (Bolded area identifies Task Standard)	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	1	**Classify the emergency event:**	**SRO classifies the event as a GENERAL EMERGENCY (GE) based on ECG – EAL Section S- System Malfunction, S1 Loss of AC Power, EAL# SG1.1 for GE.**		
*	2	**Complete the ICMF:**	**SRO accurately completes Attachment 4 Sections I thru V of the ICMF.** See attached Answer Key.		

System: Generic Admin - Emergency Plan

Task: Classify an Event and Determine PARs

* #	STEP NO.	STEP (Shaded area denotes Critical Step) (* Critical Step) (# Sequential Critical Step)	STANDARD (Bolded area identifies Task Standard)	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	2 Cont.	APPENDIX 1 - PREDTERMINED PROTECTIVE ACTION RECOMMENDATIONS	Operator uses Attachment 4, EP-SA-325-F4, Appendix 1, and correctly answers the following questions to determine PARs: ▪ Initial PAR – Yes ▪ GE based on Loss of Three Fission Product Barriers? – No ▪ Is there a Hostile Action Event in Progress affecting the GE? – No ▪ **Selects Default PAR (No RPSA)** ➢ **Evacuate All Sectors 0-5 miles** ➢ **Monitor & Prepare All Sectors 5-10 miles**		
	CUE:	**JPM is Complete** **RECORD** the STOP TIME**:** **STOP TIME:_____**	**Terminate JPM when SRO submits the ICMF form.**		

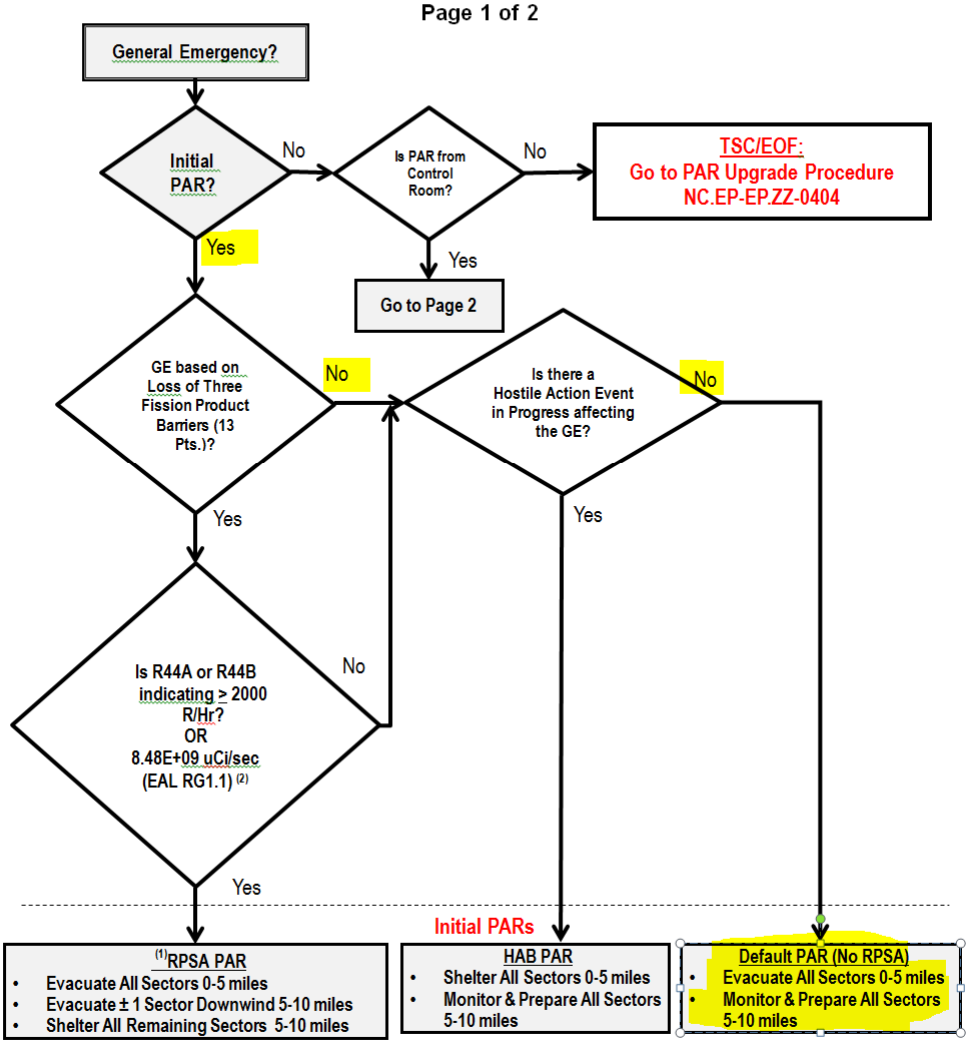
Section S – System Malfunction
S1 – Loss of AC Power

Initiating Condition	Loss of **ALL** offsite power and **ALL** onsite AC power to vital buses for 15 minutes or longer	Prolonged loss of **ALL** offsite and **ALL** onsite AC power to vital buses
MODE	1, 2, 3, 4	1, 2, 3, 4
EAL #	**SS1.1**	**SG1.1**
EMERGENCY ACTION LEVELS	**IF** Loss of **ALL** offsite and **ALL** onsite AC power to 4 KV vital buses	**IF** Loss of **ALL** offsite and **ALL** onsite AC power to 4 KV vital buses
	AND **≥ 15 minutes** have elapsed (Note 1)	**AND** **EITHER** of the following: • Restoration of at least one Vital Bus in **< 4 hrs** is **NOT** likely • CFST Core Cooling **RED** path exists (Note 1)
	THEN	**THEN**
Action Required	Refer to Attachment 3 **SITE AREA EMERGENCY**	Refer to Attachment 4 **GENERAL EMERGENCY**

Note 1: The Emergency Coordinator should declare the event promptly upon determining that time limit has been exceeded, or will likely be exceeded.

S1

APPENDIX 1
PREDETERMINED PROTECTIVE ACTION RECOMMENDATIONS
Page 1 of 2



SEE LAST PAGE OF ATTACHMENT TO DETERMINE DOWNWIND SECTORS

ANSWER KEY

INITIAL CONTACT MESSAGE FORM

I. THIS IS _____, COMMUNICATOR IN THE ☑ CONTROL ROOM
(NAME) ☐ TSC
☐ EOF

AT THE **SALEM** NUCLEAR GENERATING STATION, **UNIT**(s) No. 2

IIa. ☑ THIS IS NOTIFICATION OF AN **GENERAL EMERGENCY** WHICH WAS

DECLARED AT	Current Time (Time - 24 HR CLOCK)	ON	Today (DATE)

EAL #	SG1.1	,		,	
DESCRIPTION OF EVENT:	Prolonged Loss Of Power To All Vital Buses				

OR

IIb. ☐ THIS IS NOTIFICATION OF A PROTECTIVE ACTION RECOMMENDATION UPGRADE MADE AT _____ (Time - 24 HR CLOCK) HRS ON _____ (DATE)

Reason for PAR Upgrade:

III. ☐ THERE **IS** A RELEASE IN PROGRESS DUE TO THE EVENT
☑ THERE IS **NO** RELEASE IN PROGRESS DUE TO THE EVENT
} Any release above normal, attributable to the event. See Basis for examples

IV.	33 FT. LEVEL WIND DIRECTION (**From**): (From MET Computer /SPDS)	90 (DEGREES)	WIND SPEED:	2 (MPH)

V.	Sectors	Dist. - Mile
☑ WE RECOMMEND **EVACUATION** AS FOLLOWS	ALL	0-5
☐ WE RECOMMEND **SHELTERING** AS FOLLOWS		
☑ WE RECOMMEND **MONITOR AND PREPARE** AS FOLLOWS	ALL	5-10
☑ WE RECOMMEND THE **USE OF KI** IN ACCORDANCE WITH STATE PROCEDURES		

_____
EC Initials
(Approval to Transmit ICMF)

SGS Rev. 1

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

JPM #: 20-01 NRC SRO-A5

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

____RC__ 1. Task description and number, JPM description and number are identified.

____RC__ 2. Knowledge and Abilities (K/A) references are included.

____RC__ 3. Performance location specified. (in-plant, control room, or simulator)

____RC__ 4. Initial setup conditions are identified.

____RC__ 5. Initiating and terminating Cues are properly identified.

____RC__ 6. Task standards identified and verified by SME review.

____RC__ 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).

____RC____ 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. _____ Date _____

____RC__ 9. Pilot test the JPM:
a. verify Cues both verbal and visual are free of conflict, and
b. ensure performance time is accurate.

____NA__ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.

____RC__ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor:	R. Chan	Date	1-10-22
SME/Instructor:	M. Protesto	Date:	1-10-22
SME/Instructor:		Date:	